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UNIVERSITI TEKNOLOGI MALAYSIA

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UNIVERSITI TEKNOLOGI MALAYSIA

SECP3213-01 - BUSINESS INTELLIGENCE

INDUSTRY TALK REPORT

TITLE OF TALK:

BRIDGING DATA & DECISIONS: THE ROLE OF DATA GOVERNANCE AND DATA
MANAGEMENT IN MODERN BUSINESS INTELLIGENCE

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1.0 INTRODUCTION

The industry talk titled *“Bridging Data & Decisions: The Role of Data Governance and Data Management in Modern Business Intelligence”* was presented by Faiz Fablillah, who currently serves as the Senior Principal of Enterprise Data & AI Governance at Perbadanan Insurance Deposit Malaysia and is also a board member of DAMA Kuala Lumpur & Selangor. The speaker gave us many useful examples from his long experience in various industries, such as financial services, manufacturing and oil and gas. The primary goal of the talk was to highlight the need for Data Governance and Data Management in making the Business Intelligence (BI) system useful for accurate and reliable business decisions. The session focused on the fact that it is important not only to make visually appealing dashboards, but also to ensure that the data that is displayed behind the scenes is accurate, consistent, secure and properly managed. The talk is highly relevant to Business Intelligence and Analytics because the organizations today rely on data-driven strategies, artificial intelligence, and automated systems to optimize their operations and stay competitive in the digital era.

2.0 KEY TAKEAWAYS

One of the major points highlighted during the talk was the growing importance of analytics engineering in connecting raw data with meaningful business decisions. The speaker discussed how a lot of organizations invest in dashboards and visualization tools, but fail to invest in the quality and governance of the data. There were multiple industry statistics presented to illustrate this, with a significant proportion of enterprise data being unused and a poor quality data impacting can lead to significant financial consequences for organisations. The speaker also articulated that data pipelines in real-world organizations are generally very complex, as various departments use and store data in different ways. This makes data engineers often find themselves struggling with problems like data asset ownership, lack of documentation, shifting database schemas, and data inconsistencies. The focus of the talk was on the importance of effective communication and collaboration between departments, to ensure that data pipelines are reliable and effective.

Another key learning during the session was the discussion around the data quality dimensions and how it affects business operations. The speaker provided a number of examples of data quality problems in real-world scenarios that can have a detrimental impact on organizations. For instance, legacy customer information could result in banks sending out sensitive financial offers to deceased individuals, and faults in data pipelines can result in incorrect reporting within health care systems. The speaker explained to the audience that there were three aspects of data quality that were important: accuracy, timeliness, and consistency. These dimensions are very important since business leaders depend heavily on dashboards and analytical reports for their strategic decision making. Inaccurate or inconsistent data can lead to a lack of trust in the reporting

processes and result in suboptimal business decisions. Hence, data governance frameworks and automated data quality controls are gaining relevance in today's businesses.

The talk also delved into the use of artificial intelligence (AI) in enhancing data governance and management processes. The speaker illustrated how manual monitoring of enterprise scale data systems becomes inefficient and hard to maintain as the volume of data increases faster and faster. AI-powered anomaly detection systems are now being adopted by modern organizations to detect anomalies like missing values, unexpected trends, and corrupt datasets before they impact dashboards or machine learning systems. Another area is the processing of unstructured data, like images, scanned documents and raw text files, where AI technologies are playing a role. Organizations can transform unstructured data into search-able and machine-readable formats using methods like Retrieval-Augmented Generation (RAG) and vector databases. The speaker then spoke about the Modern Data Platform, which is a platform that encompasses ingestion, processing, storage, governance, and analytics. In this context, Business Intelligence is becoming Generative Business Intelligence (GenBI), where people who are not technically savvy can use natural language to ask questions of an AI-based system for the automatic generation of dashboards and insights, rather than having to create them manually by using technical skills.

3.0 REFLECTION

This talk was very informative on the real issues dealing with organizations while implementing Business Intelligence systems. Some of the most intriguing topics raised were the connection between Data Engineering and Data Governance. Data engineers typically are more interested in speed and efficiency to deliver data rapidly, and governance teams are more interested in validation, security, compliance, and data quality. Although these objectives may sometimes conflict, the speaker explained that both areas are equally important, especially with the increasing use of AI and machine learning technologies. This discussion clearly relates to topics covered in the SECP3213 Business Intelligence course where students are introduced to structured ETL processes, dashboards and data visualization techniques. The talk, however, let it be known that “real-world” enterprise data environments are more complex and less organized than the examples in class. It emphasized the need to have technical analytics skills, as well as an understanding of metadata, data lineage, ownership, and governance frameworks.

In addition, the presentation shifted my mindset on the function of analytics engineering in today's businesses. My previous perspective on analytics engineering was primarily as a technical person who deals with SQL queries, dashboards, and data transformation processes. But the speaker showed that analytics engineers should also have communication skills, business acumen, and knowledge about governance practices to guarantee accurate and impactful analytical results for business decision makers. This session also made me think more deeply about data management within University projects as well as my future career. The project should also involve documentation, validation and data quality management, not just to create

eye-catching dashboards. These practices are crucial because regardless of how advanced visualization tools and AI models are, if the data is not managed properly, they will not guarantee reliable insights.

4.0 CONCLUSION

The industry talk by Faiz Fablillah was a valuable and practical one, where it has been clearly understood the significance of data governance and data management in the current day business Intelligence systems. It was evident during the session that robust analytics and AI efforts require reliable, accurate, and effective data infrastructures that are well managed. Without adequate data management and validation, organizations shouldn't expect advanced dashboards or machine learning technologies to come in handy. The presentation also showcased the latest advancements in AI-driven data quality, anomaly detection, and the management of unstructured data. In general, this session has shaped my knowledge of Business Intelligence and my academic and career choices in the direction of Data Governance, Analytics Engineering and Enterprise Data Management. It also emphasized the need to have both technical knowledge and understanding of governance and business to develop data-driven solutions in the future which are reliable and sustainable.